2026
Practical Parallel Computing
(実践的並列コンピューティング)

Part 3: CUDA

No. 2: Using Many Threads in CUDA

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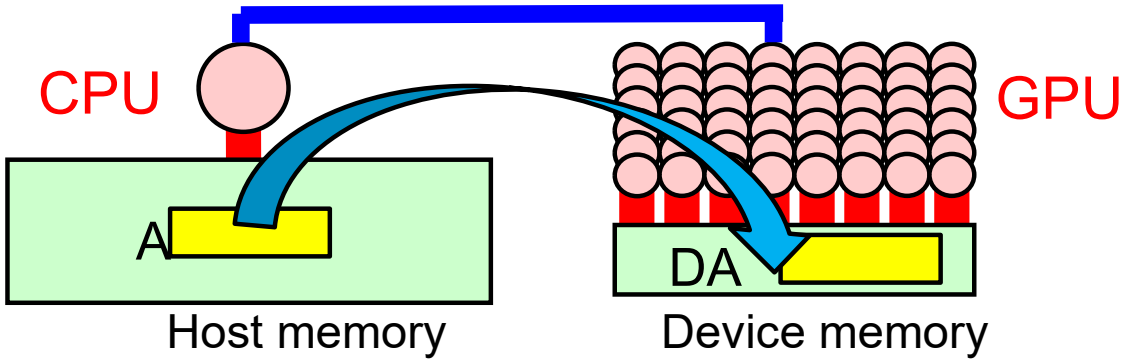
Overview of This Course

- Part 0: Introduction
  - 2 classes
- Part 1: Using OpenMP (OMP)
  - 4 classes
  - Report (required)
- Part 2: GPU programming with OpenACC (ACC)
  - 2 classes
  - Report (required)
- Part 3: GPU programming with CUDA ← We are here (2/3)
  - 3 classes
  - Report (elective)
- Part 4: Using MPI
  - 3 classes
  - Report (elective)

Review: How We Write CUDA Programs

- Data is prepared on GPU device memory
  - cudaMalloc(), cudaMemcpy()



- CPU function calls GPU kernel functions

[CPU side]

```
func<<<20, 5>>>(…);
```

In this case, 20x5=100 threads run on GPU

call

return

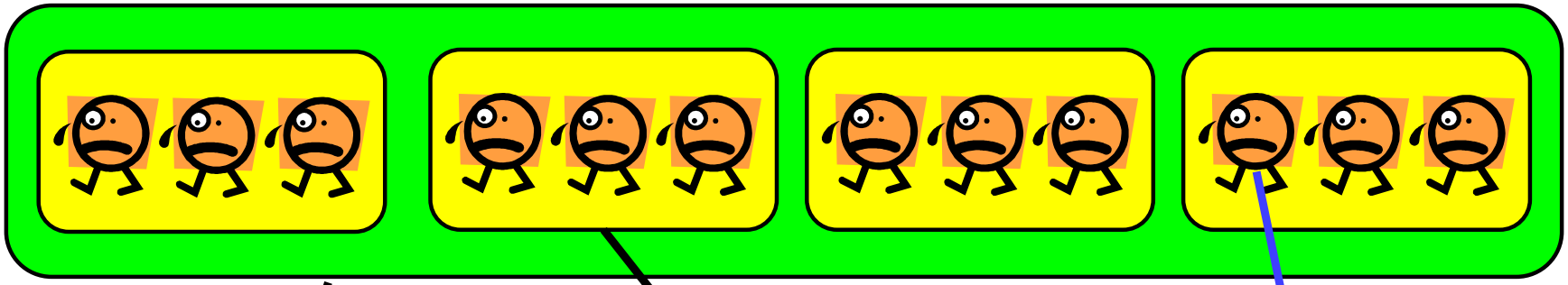
[GPU side]

```
__global__ void func(…)
{
        :
  return;
}
```

Review:
Grid, Thread Blocks, Threads

A grid has several thread blocks

A thread block has several threads



A grid

A thread block

A thread

cf) func <<< 4, 3 >>> (); → 12 threads

Number of thread blocks
= gridDim

Number of threads per block
= blockDim

Review:
How is Number of Threads Designed?

We have to deicide 2 numbers

<<<block number, block size>>>

A better way would be

(1)We decide total number of threads P

- P should be very large, such as >1,000,000

(2)We tune each block size BS

- Good candidate are 16, 32, 64, … 1024

(3)Then block number is P/BS

- It is better to consider indivisible cases

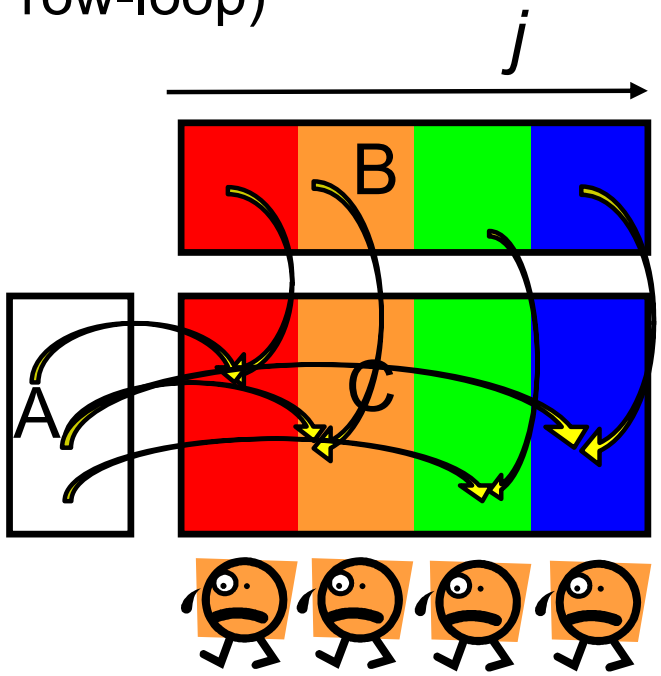
Review: mm-1dpar Sample is Slow

Related to [C3]

CUDA versions of mm samples are at
- ppcomp-ex/cuda/mm-1dpar/ (slow)
- ppcomp-ex/cuda/mm/ (faster)

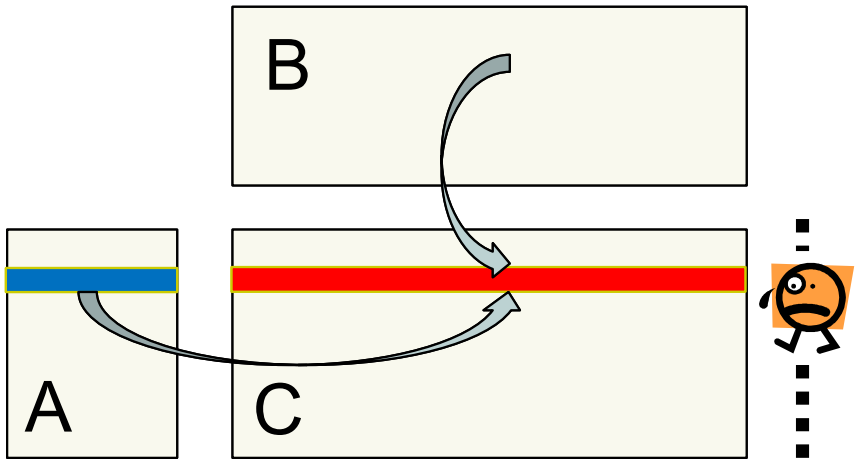
OpenMP

- Parallelize column-loop (or row-loop)



CUDA (mm-1dpar)

- We can create many threads
- 1 thread computes 1 row
  - We use m threads



m threads are too few for GPU

In Case of mm-acc (OpenACC)

- mm use “acc loop independent” twice
  - j loop (size n) and i loop (size m)

```
#pragma acc data copyin(A[0:m*k],B[0:k*n]),copy(C[0:m*n])
#pragma acc kernels
#pragma acc loop independent   // j loop is parallelized
  for (j = 0; j < n; j++) {
#pragma acc loop seq
    for (l = 0; l < k; l++) {
#pragma acc loop independent    // i loop is parallelized
      for (i = 0; i < m; i++) {
        Ci,j += Ai,l * Bl,j;
      } } }
```

➔ This program creates m * n threads internally

➔ Also we want to create m * n threads on CUDA

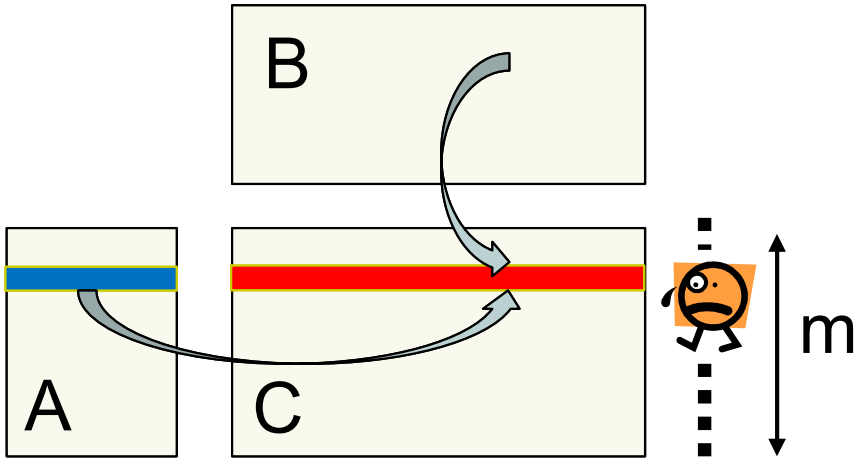
Improvement of mm: How to Use More Threads

In mm, computation of each C element is independent with each other

cuda/mm-1dpar

- 1 thread computes 1 row

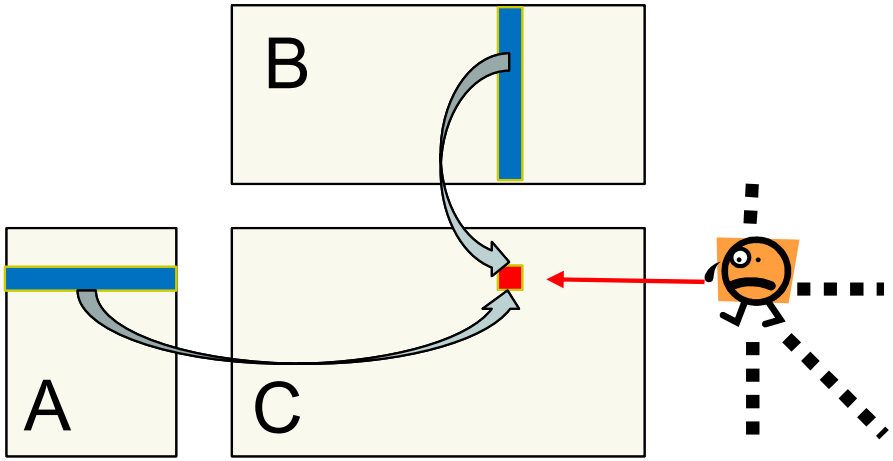
➔ We use m threads



cuda/mm

- 1 thread computes 1 element

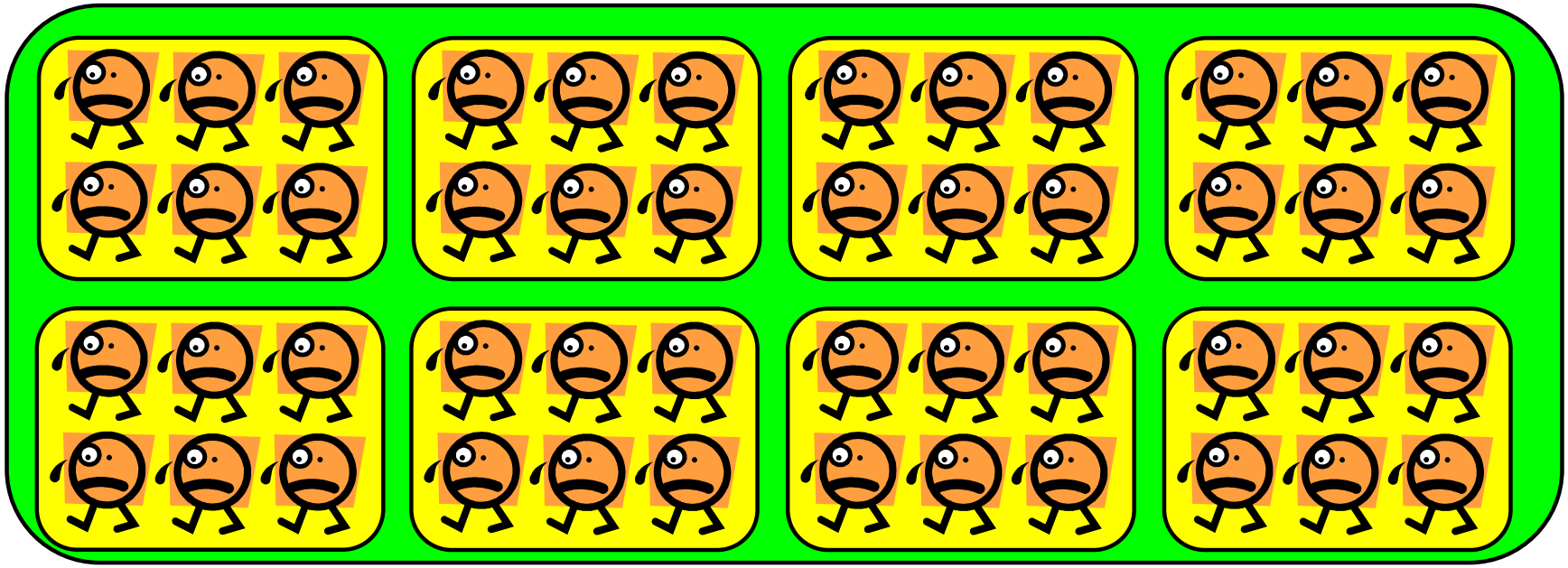
➔ We use m × n threads !!



Creating Threads with 2D/3D IDs

- Now we want to make m*n (may be >1,000,000) threads
  - <<<(m*n)/BS, BS>>> is ok, but coding is bothersome
- On CUDA, thread blocks and thread can have 3D IDs
  - gridDim and blockDim may have "dim3" type, 3D vector structure with x, y, z fields

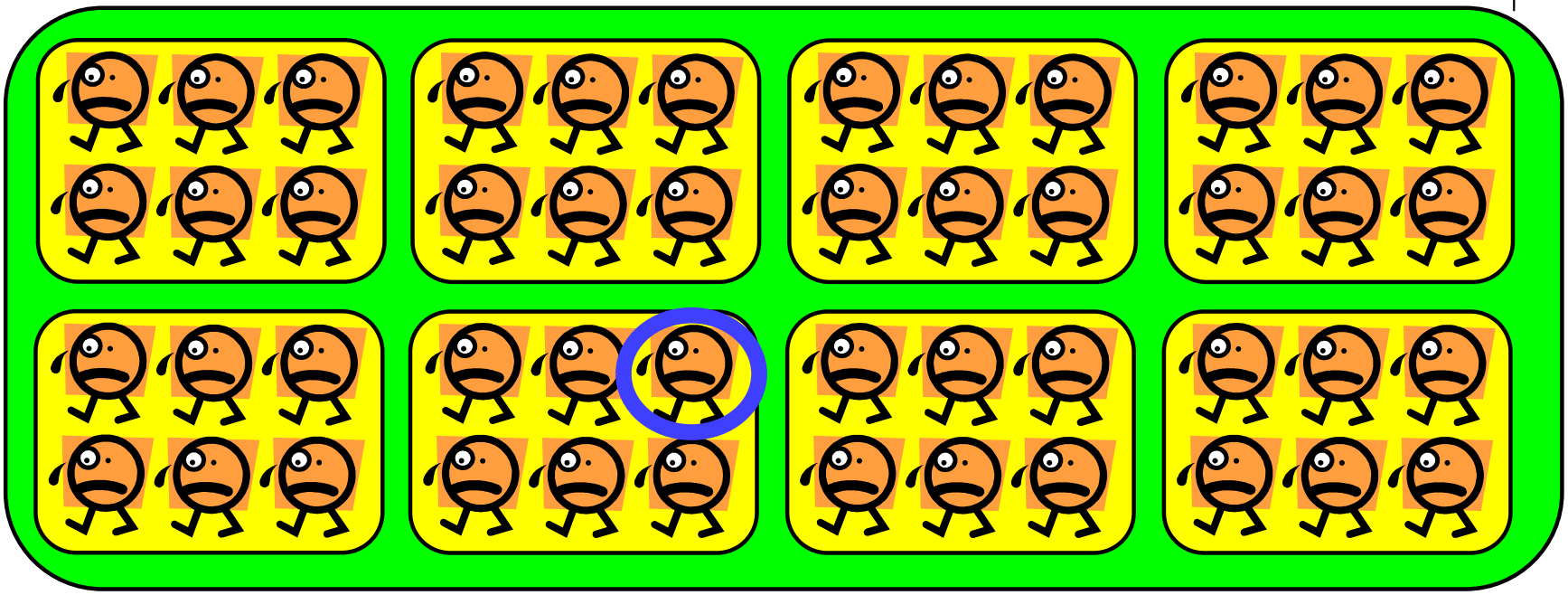
cf) func <<< dim3(4,2,1), dim3(3,2,1) >>> (); → 48 threads



※ This example is the case of 2D (Z dimensions are 1)

Thread IDs in multi-dimensional cases

In the case of func <<< dim3(4,2,1), dim3(3,2,1) >>> ();



- For every thread,
  gridDim.x=4, gridDim.y=2, gridDim.z=1
  blockDim.x=3, blockDim.y=2, blockDim.z=1
- For the thread with blue mark,
  blockIdx.x=1, blockIdx.y=1, blockIdx.z=0
  threadIdx.x=2, threadIdx.y=0, threadIdx.z=0

Notes on 1D Specification

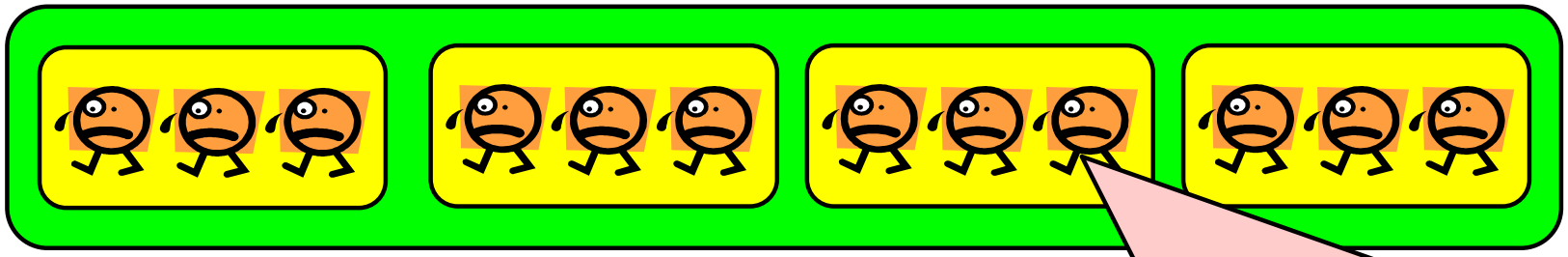
- So far we used 1D gridDim, 1D blockDim
- They are special cases of 3D IDs

The followings are same meanings

```
func <<<4,  3>>> ();
```

```
func <<<dim3(4,1,1),  dim3(3,1,1)>>> ();
```

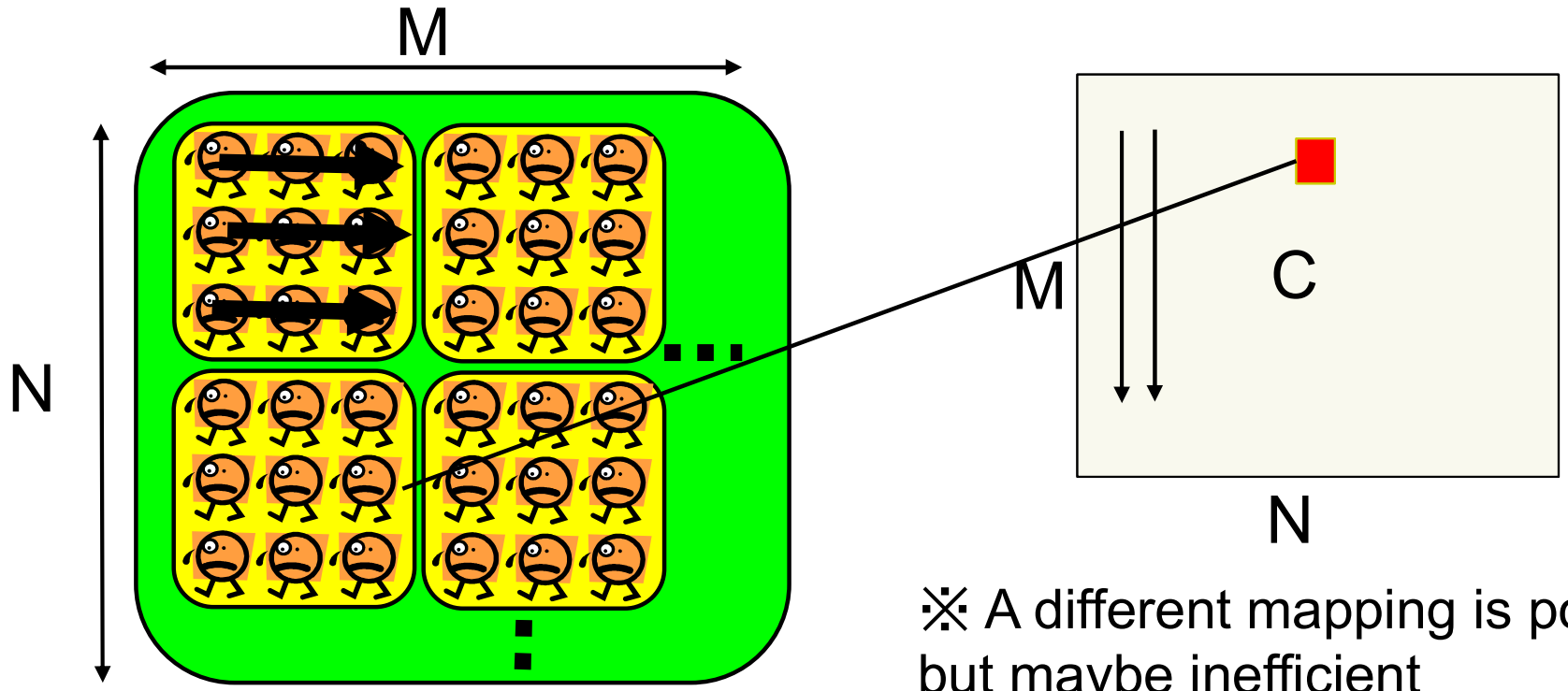
```
dim3 grid = dim3(4,1,1);
dim3 block = dim3(3,1,1);
func <<<grid,  block>>> ();
```



blockIdx.x = 2, blockIdx,y = 0, blockIdx.z = 0
threadIdx.x = 2, threadIdx.y = 0, threadIdx,z = 0

Using More Threads in cuda/mm Sample

- The total number of threads should be m*n
- How do we determine gridDim, blockDim?
  - <<<m, n>>> does not work for CUDA constraints
- Here, we determine blockDim as x=16, y=16 → 256 threads per block
  - Then gridDim is computed from M, N
- x is mapped to row index, y is mapped to column index (※)



※ A different mapping is possible, but maybe inefficient

CPU Side Code in cuda/mm

gridDim blockDim

```
matmul_kernel<<<dim3(m / BS, n / BS, 1), dim3(BS, BS, 1)>>>
    (DA, DB, DC, m, n, k);
```

BS=16 in this sample

Using rounding up for indivisible cases...

```
dim3 grid = dim3((m+BS-1)/BS, ((n+BS-1)/BS), 1);
dim3 block = dim3(BS, BS, 1);
matmul_kernel<<<grid, block>>>(DA, DB, DC, m, n, k);
```

GPU Side Code in cuda/mm

```
__global__ void matmul_kernel(double *DA, double *DB, double *DC, int m, int n, int k)
{
    int i, j, l;
    j = blockIdx.y * blockDim.y + threadIdx.y;
    i = blockIdx.x * blockDim.x + threadIdx.x;
    if (i >= m || j >= n) return; // do nothing

    double cij = DC[i+j*ldc];
    for (l = 0; l < k; l++) {
        cij += DA[i+l*m] * DB[l+j*n];
    }
    DC[i+j*ldc] = cij;
}
```

A thread gets
entire thread IDs (x,y)
j ← y dimension, i ← x dimension

Avoid too many threads

A thread computes C_{ij}
➔ Only l loop (dot prod)

Notes about Report [C3]

- Compare speeds of
  - cuda/mm-1dpar: m threads
  - cuda/mm: m*n threads
  - Please use various matrices sizes (m, n, k)
- Evaluate effect of data transfer cost
  - The sample prints measured time in detail
  - Discussions are similar to OpenACC version (2-2 slides)
    - Computation complexity: O(mnk)
    - Data transfer complexity: O(mk+kn+mn)

CUDA Rules on Number of Threads

func<<<A, B>>> (...); *(A, B are integers)* is same as
func<<<dim3(A,1,1), dim3(B,1,1)>>> (...);

gridDim blockDim

func<<<dim3(gx, gy, gz), dim3(bx, by, bz)>>> (...);

$\leqq 2^{31}-1$

$\leqq 65535$

$\leqq 1024$

$\leqq 64$

Also, bx*by*bz must be $\leqq 1024$

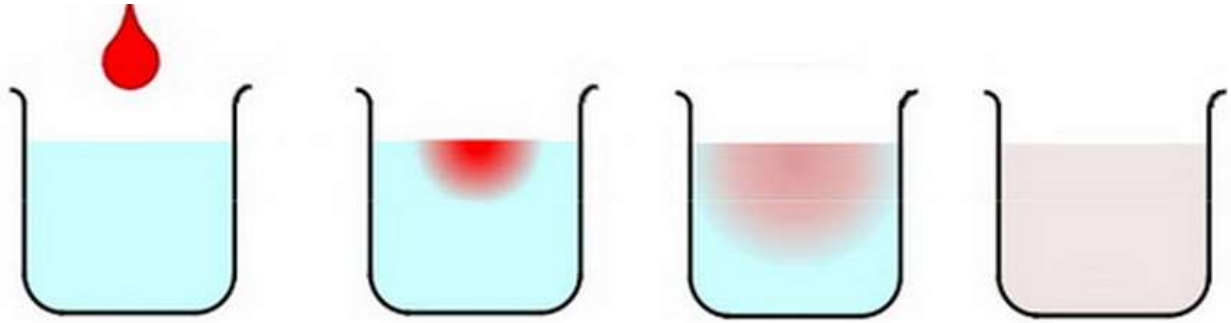
Size of BlockDim has severe limitation ☹

Cf) <<<m, n>>> causes an error if n>1024 ☹

“diffusion” Sample Program

Target of [C1] , details are in 1-2 slides

An example of diffusion phenomena:

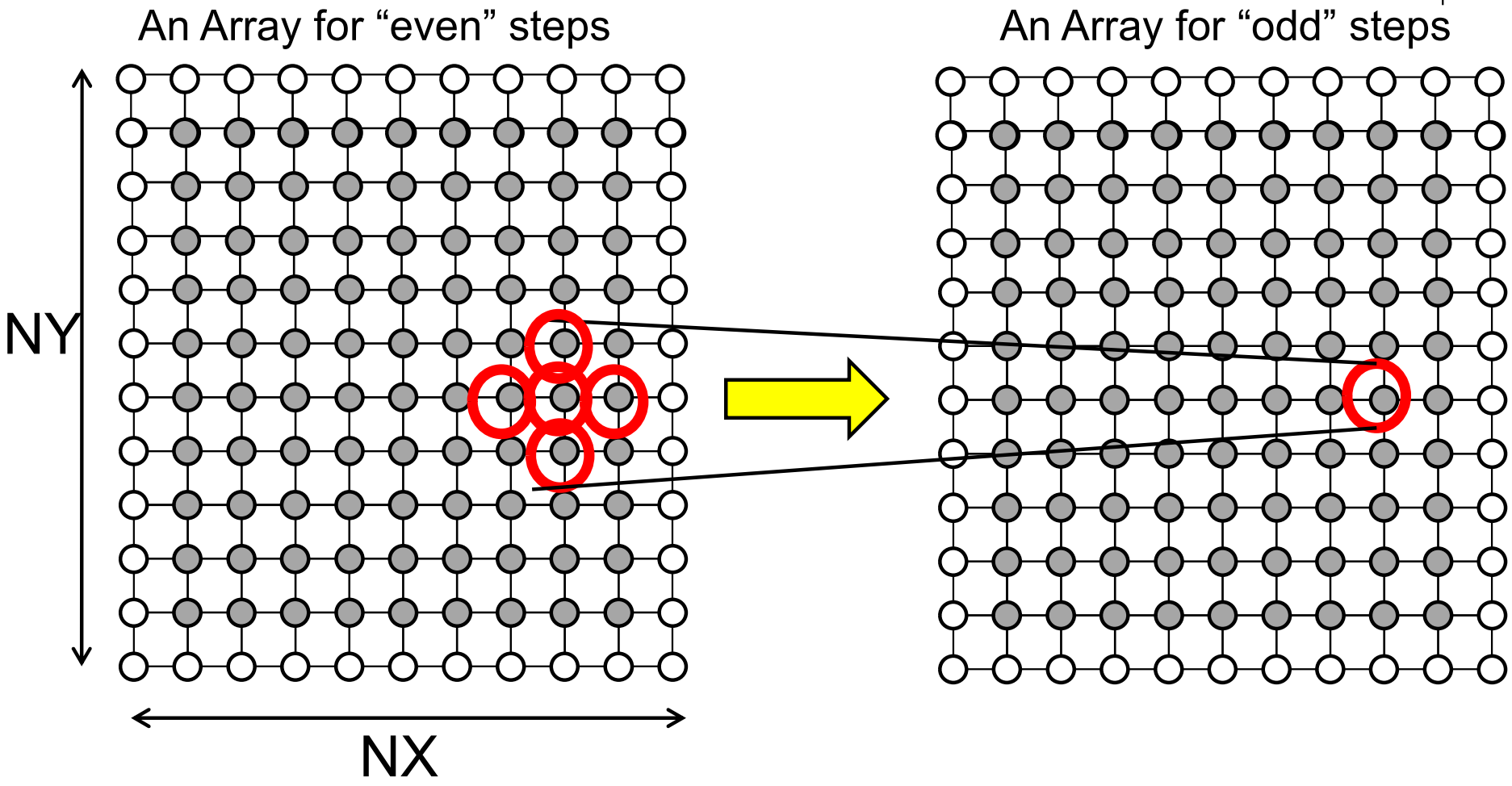


The ink spreads gradually, and finally the density becomes uniform (Figure by Prof. T. Aoki)

Base version: ppcomp-ex/base/diffusion/
You can use ppcomp-ex/cuda/diffusion/

```
cd ppcomp-ex/cuda/diffusion
module load nvhpc
make
./diffusion 20      // number of time steps
```

Discussion on diffusion sample



Both arrays have to be on GPU device memory when computations are done

Consideration of Parallelizing Diffusion

- x, y loops can be parallelized
- t loop cannot be parallelized

```
[Data transfer from CPU to GPU]
for (t = 0; t < nt; t++) {

  for (y = 1; y < NY-1; y++) {
    for (x = 1; x < NX-1; x++) {
          :
    }
  }
}
[Data transfer from GPU to CPU]
```

GPU computation must be a distinct function (GPU kernel function)

It's better to transfer data *out of* t-loop

Preparing GPU Kernel Function

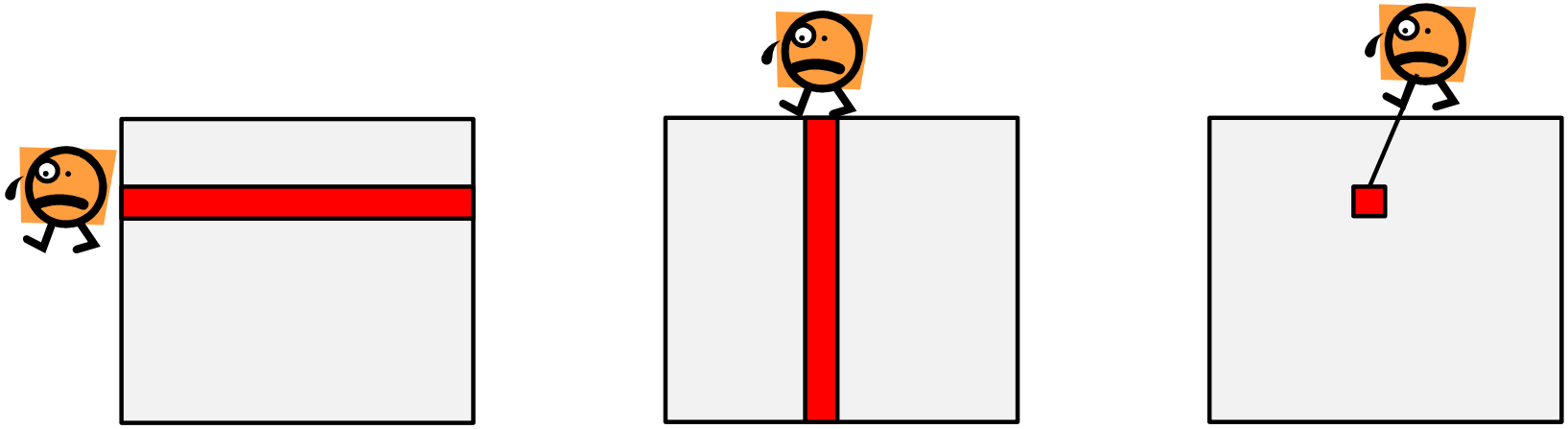
- Unlike OpenACC/OpenMP, region on GPU must be a separated function. Code needs large change!

```
__global__ void calc_gpu(...) {
    :   // behavior of each thread
}

int calc(int nt) {
    :
  for (t = 0; t < nt; t++) {
    :
    calc_gpu<<<....>>>(...)
    :
  }
}
```

Calling a kernel function

Considering CUDA Threads

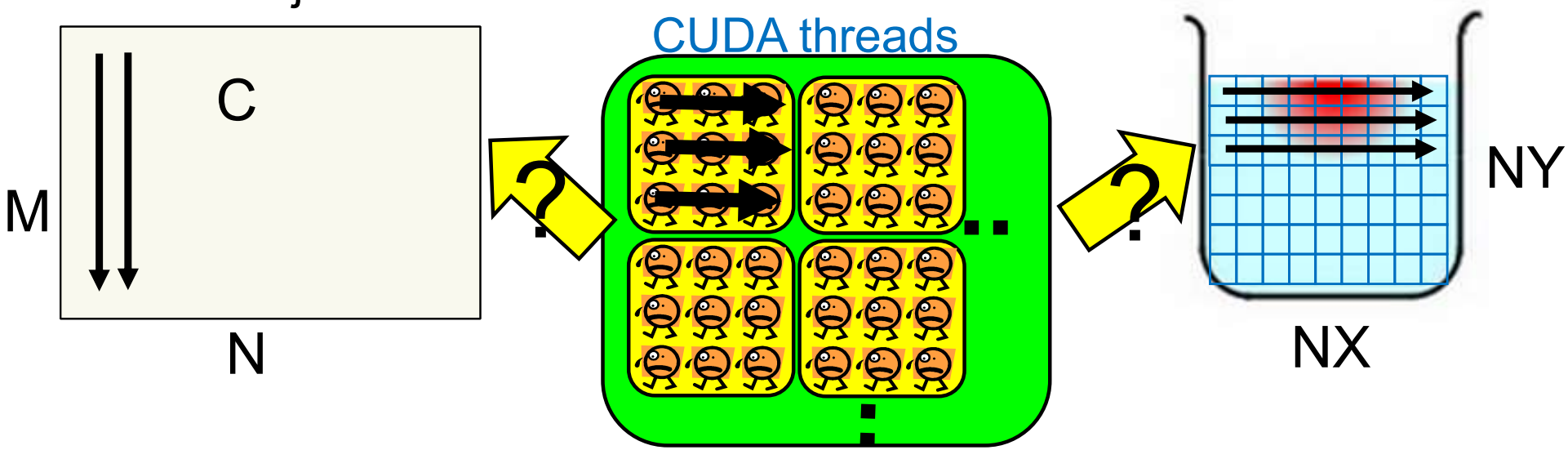
- How do we design threads on CUDA?
- There are several choices
  - 1thread = 1row
    - We use NY threads in total → only x-loop in kernel function
  - 1thread = 1column
    - We use NX threads in total → only y-loop in kernel function
  - 1thread = 1element (Recommended)
    - We use NX x NY threads in total → No loop in kernel function!
    - This will be fast since the number of threads is very large



Mapping between Threads and Data

mm:
Matrices has
column-major format

diffusion:
2D array has
row-major format




```
j = blockIdx.y * blockDim.y +
threadIdx.y;
i = blockIdx.x * blockDim.x +
threadIdx.x;
 : This thread computes Cij
```

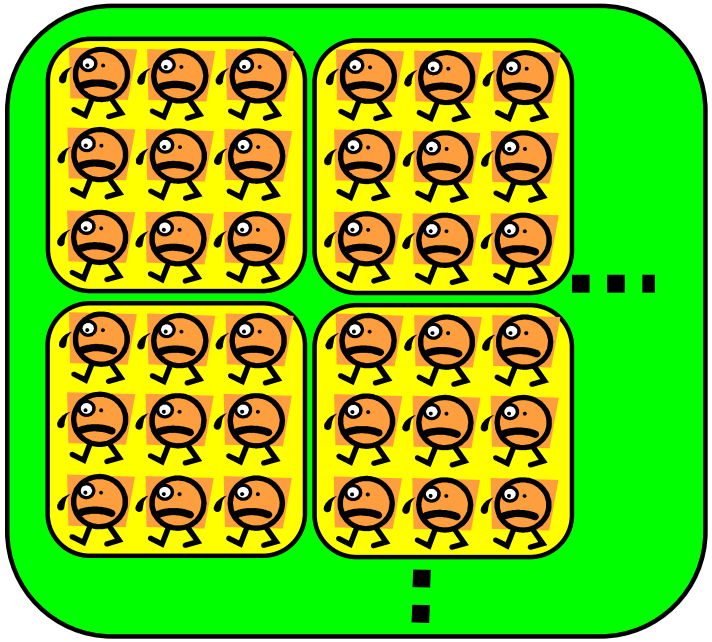
```
y = blockIdx.y * blockDim.y +
threadIdx.y;
x = blockIdx.x * blockDim.x +
threadIdx.x;
 : This thread computes[y][x]
```

[Q] What if the dimensions are exchanged?

Considering gridDim/blockDim (1)

func <<< dim3 (?, ?, ?), dim3 (?, ?, ?) >>>(…);

gridDim blockDim



(1) We decide total number of threads

➔ (NX, NY, 1) threads

- See notes on the next page

(2) We tune each block size (blockDim)

➔ Good candidates are (4, 4, 1), (8, 8, 1), (16, 16, 1), (32, 32, 1)

- The number must be ≦ 1024
- How about non-square blocks?

(3) Then block number (gridDim) is determined

We should consider indivisible cases

Considering gridDim/blockDim (2)

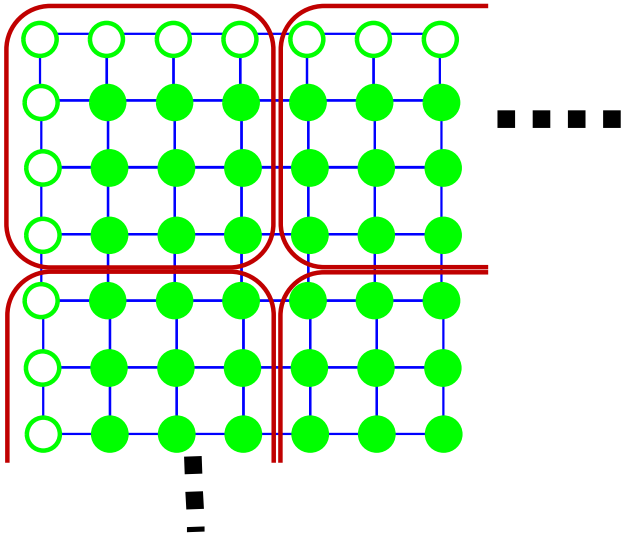
- In diffusion, Points [1, NX-1) × [1, NY-1), <u>excluded boundary</u>, should be computed

  There are choices:

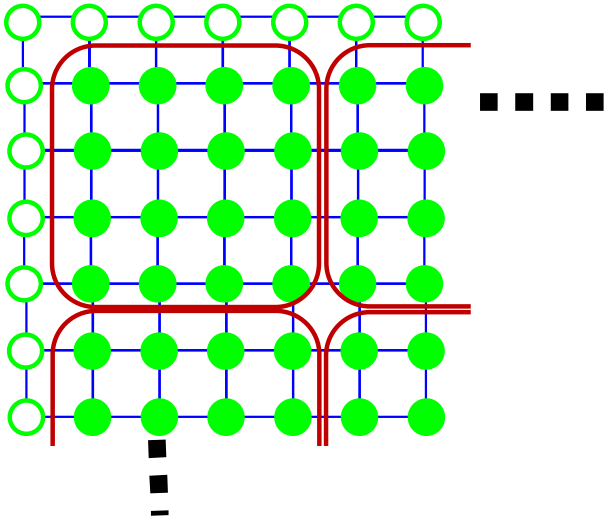
  (A) Create NX x NY threads
  - Thread (x,y) computes (x,y)
  - Threads with below IDs do nothing
    - x == 0 or y == 0 or x ≧ NX-1 or y ≧ NY-1

  (B) Create (NX-2) x (NY-2) threads
  - Thread (x,y) computes (x+1,y+1)
  - Threads with below IDs do nothing
    - x ≧ NX-2 or y ≧ NY-2

(A)



(B)



Either is ok ☺

Discussion on Data Transfer of Diffusion

Both codes will work, but how about speeds?

```
[Data transfer from CPU to GPU]
for (t = 0; t < nt; t++) {
            :

  for (y = 1; y < NY-1; y++) {
    for (x = 1; x < NX-1; x++) {
            :
    }
  }

}
[Data transfer from GPU to CPU]
```

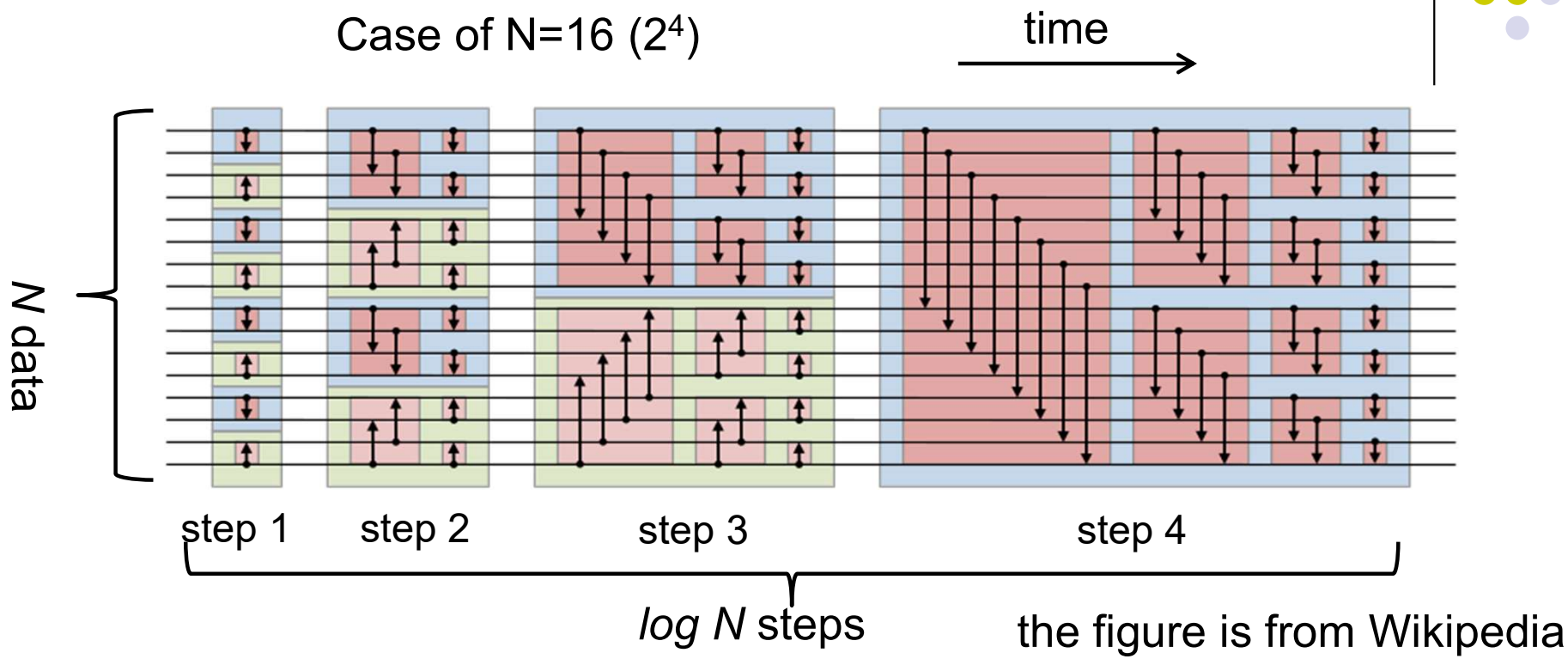
Computation: O(NX NY nt)
Transfer: O(NX NY)

```
for (t = 0; t < nt; t++) {
            :
  [Data transfer from CPU to GPU]
  for (y = 1; y < NY-1; y++) {
    for (x = 1; x < NX-1; x++) {
            :
    }
  }
  [Data transfer from GPU to CPU]
}
```

Computation: O(NX NY nt)
Transfer: O(NX NY nt)

“bsort” Sample Program

Target of [C2], details are in 2-2 slides



the figure is from Wikipedia

Base version: ppcomp-ex/base/bsort/
You can use ppcomp-ex/cuda/bsort/

```
cd ppcomp-ex/cuda/bsort
module load nvhpc  // if not yet
make
./bsort 1000000 // number of elements to be sorted
```

How Can We Parallelize bsort?

- k loops can be parallelized
- i, j loops cannot be parallelized

```
for (i = 1; (1<<i) <= N2; i++) { // step loop
  for (j = i-1; j >= 0; j--) { // sub-step loop
    :
    for (k = 0; k < N2; k++) {
        // compare 2 data and swap
    }
  }
}
```

GPU computation must be a distinct function (GPU kernel function)

Next, where should the cudaMemcpy be?

- Please consider reducing data copy cost

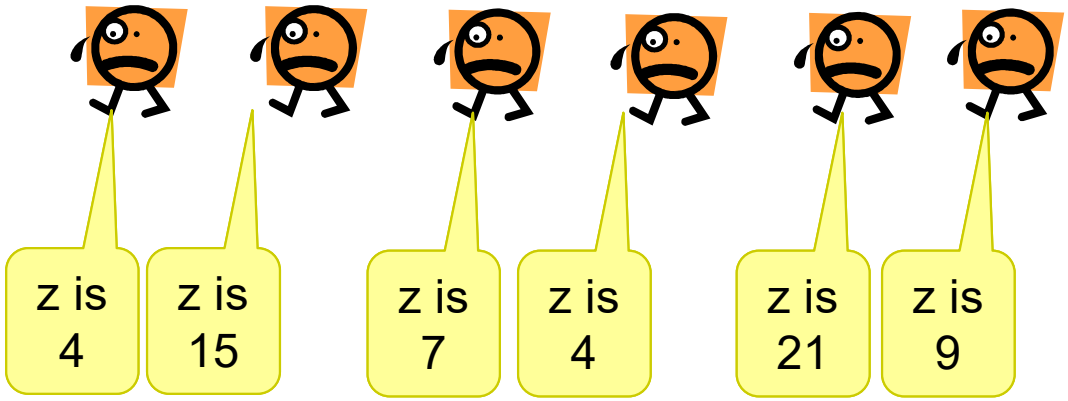
Preparing GPU Kernel Function for bsort

```
__global__ void sort_kernel (...) {
  int k = blockIdx.x * blockDim.x
      + threadIdx.x;
    :
  // behavior of each thread
}

int sort(...) {
    :
  for (i = 1 ...) {
    for (j = i-1 ...) {
      :
      sort_kernel<<<....>>>(...)
      :
    } ...
```

Calling a kernel function
The total number of threads should be N2

Rules for Memory/Variables

- Variables declared in GPU kernel functions are "thread private"

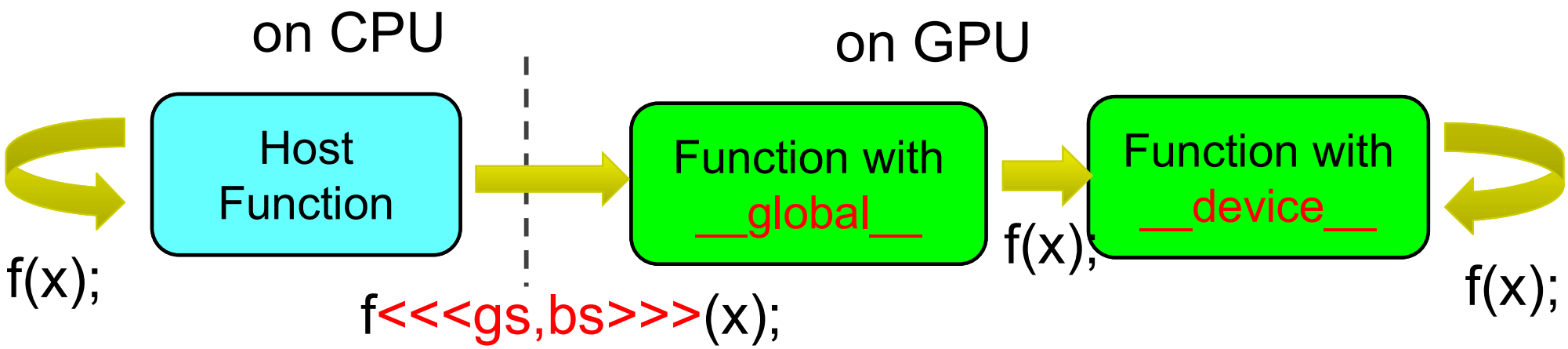


- Device memory is shared by all CUDA threads
  - Be careful to avoid race condition problem (multiple threads write same address)
  - Reading same address is ok
- Do not forget host memory and device memory are separated

Two Types of GPU Kernel Functions

1) Functions with __global__ keyword
   - "Gateway" from CPU
   - Return value type must be "void"
2) Function with __device__ keyword
   - Callable only from GPU
   - Can have return values
   - Recursive call is OK

→ In OpenACC, #pragma acc routine



What Can be Done in GPU Functions?

- Basic computations (+, -, *, /, %, &&, ||...) are OK
- if, for, while, return are OK
- Device memory access is OK
- Host memory access is NG
- Calling host functions is NG
- Calling most of functions in libc or other libraries for CPUs are NG
  - Several mathematical functions, sin(), sqrt()… are OK
  - printf() is OK
  - Calling malloc()/free() on GPU is OK, if the size must be small
    - Usually, use cudaMalloc() on <u>CPU</u>

pi sample in CUDA: Download

- pi sample in CUDA is newly added to github repository
- Please update your files as follows

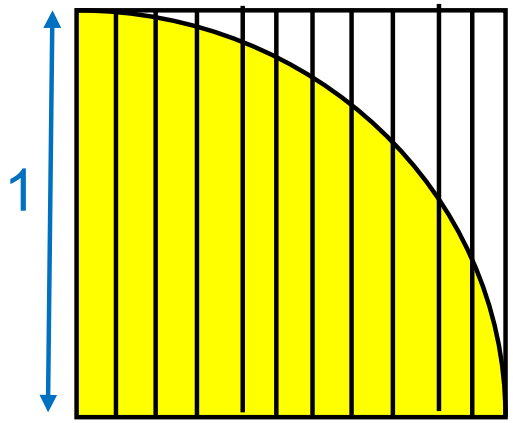
```
[please go to your ppcomp-ex directory]
git pull

[Please check if cuda/pi directory has been made]
cd cuda/pi
```

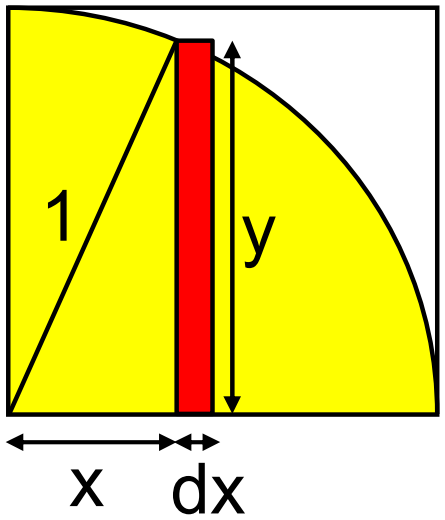
“pi” sample

Compute an approximation of π = 3.14159…
(circumference/diameter)

- ppcomp-ex/base/pi/
- Method
  - SUM ← Approximation of the yellow area
  - π ← 4 x SUM
- Execution : ./pi [n]
  - n: Number of division
  - Cf) ./pi 100000000
- Compute complexity : O(n)

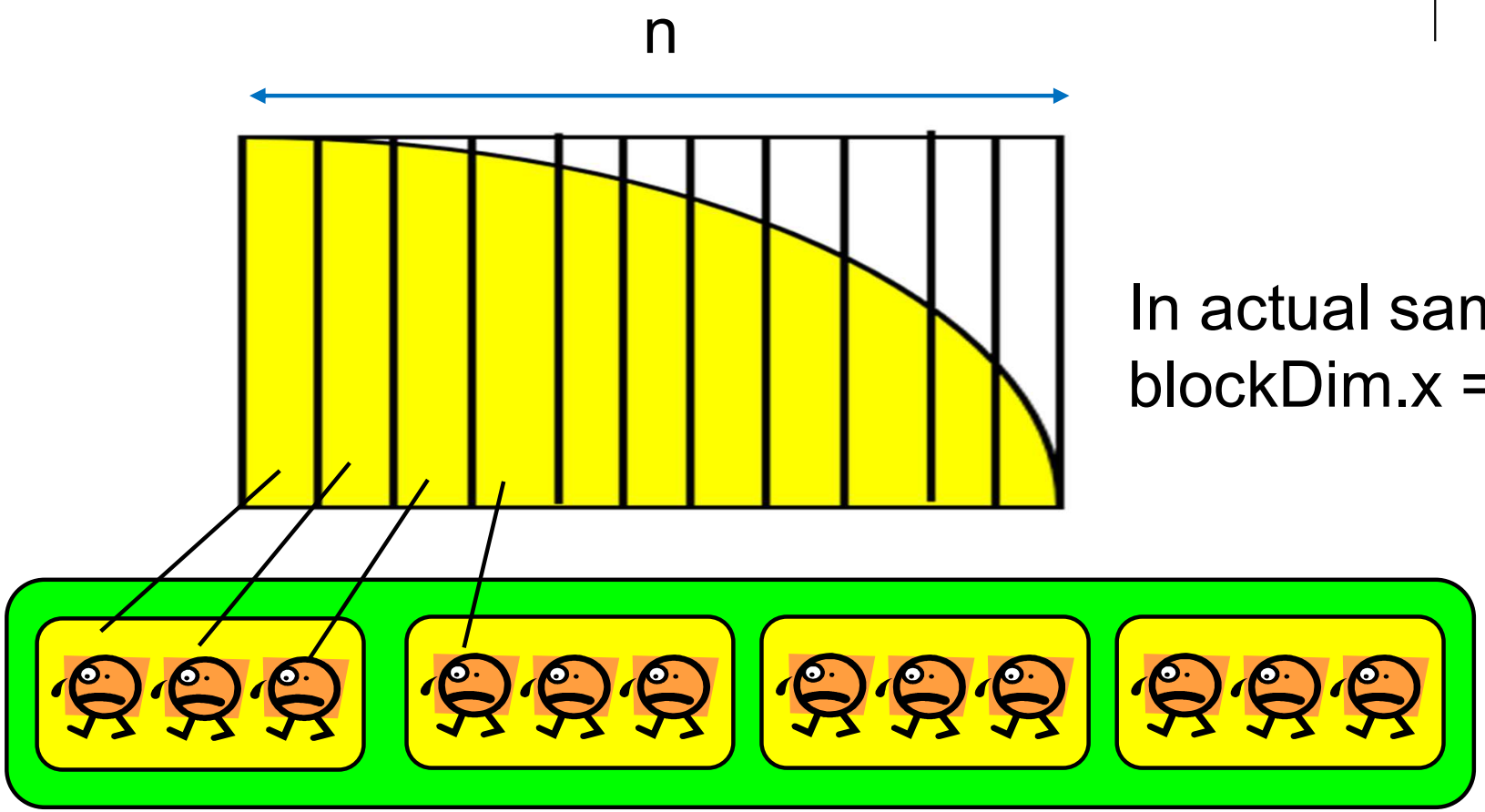
Quarter circle





dx = 1/n
y = sqrt(1-x*x)

pi Sample in CUDA



In actual sample,
blockDim.x = 256

- (Roughly) n threads are invoked, and 1 thread computes 1 partition
- Then we compute the sum ➔ Reduction is not easy in CUDA

GPU Functions in This Sample

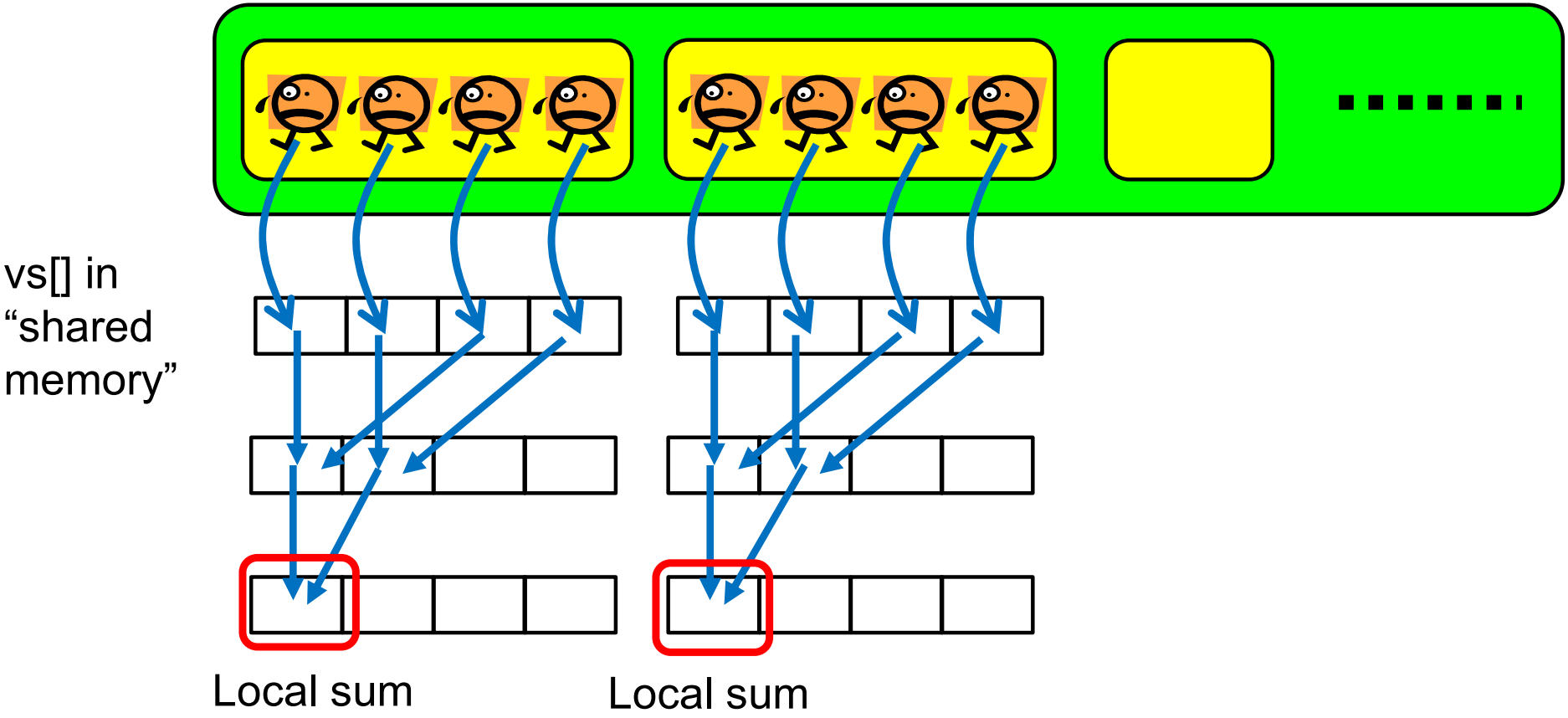
```
__global__ void pi_kernel(…)
{

   // compute my answer
        :
   v = dx*y;

   // summation!
   calc_sum(dp, n, v);
}
```

```
__device__ void calc_sum(…)
{
   // step 1: sum in each block
        :

   // step 2: the final sum
   if (tid == 0) {
        atomicAdd(dp, vs[0]);
   }
}
```

- Function call is used
  - __global__: GPU functions called by CPU
  - __device__: GPU functions called by GPU

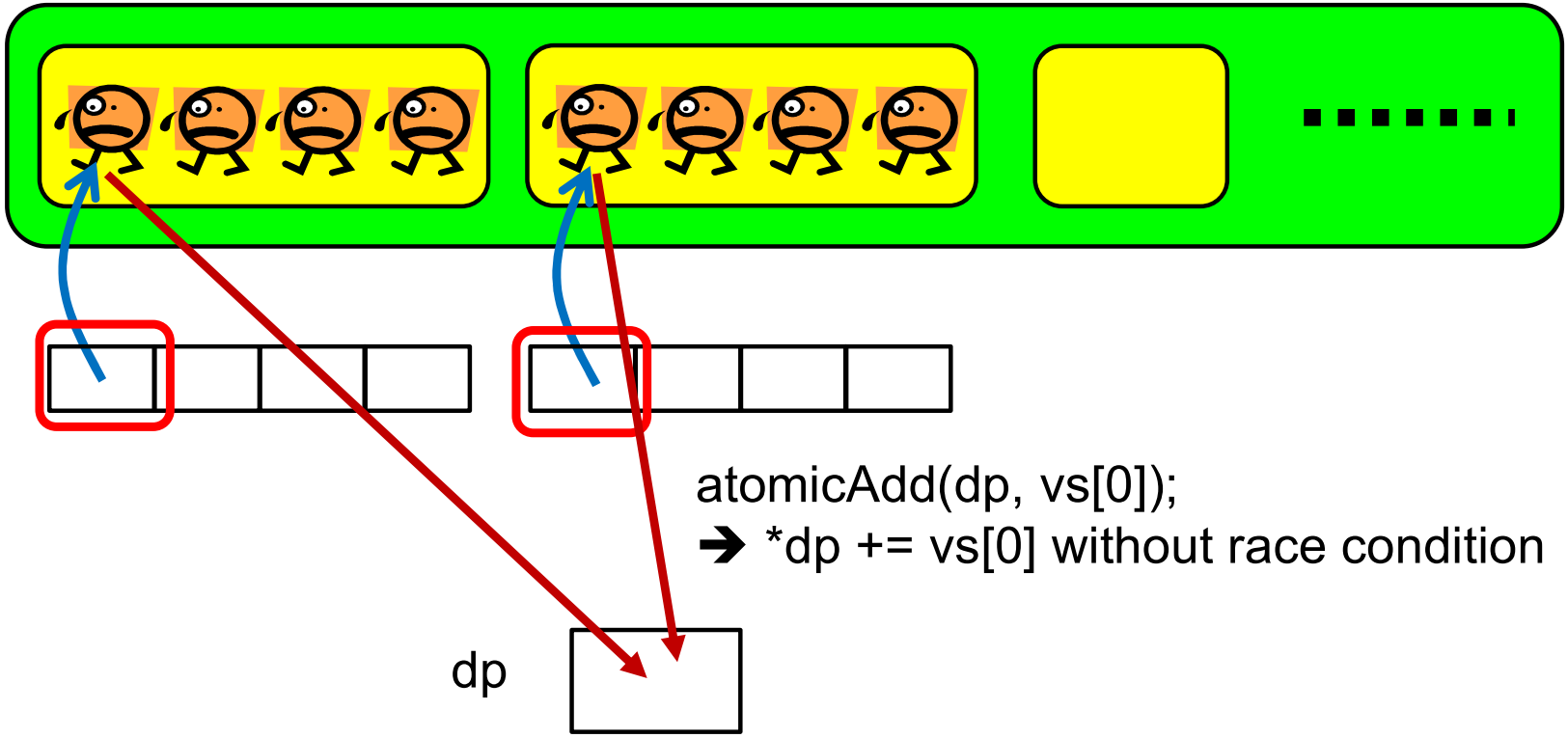
Reduction Step 1

- Computes a sum in each block
  - Fast “shared memory” is used (next class)
  - __shared__ vs[BS] in code



Reduction Step 2

- Thread 0 in each block works
- It adds local sum to the final result, *dp
  - atomicAdd is used to avoid race condition



Notes in Synchronization

- Thread synchronization: Each thread waits for other threads reach the same point
  - OpenMP on CPU: #pragma omp barrier

- On GPU, synchronization in the same thread block is supported
  - __syncthreads(): see calc_sum() in cuda/pi sample
- How about all-threads synchronization?
  - No simple generic method ☹

➔ The best way may be to return from GPU function to CPU

Assignments in CUDA Part

Choose one of [C1]—[C4], and submit a report

Due date: May 28 (Thursday)

[C1] Parallelize “diffusion” sample program by CUDA

[C2] Parallelize “bsort” sample program by CUDA

[C3] Evaluate speed of “cuda/mm” sample in detail

[C4] (Freestyle) Parallelize *any* program by CUDA

For more details, please see ppcomp25-3-1 slides

Plan of CUDA Part

- No. 1
  - Introduction to CUDA, kernel functions
- No. 2 (Today)
  - Characteristics of grid, thread blocks, threads
- No. 3
  - Performance improvement on GPU